15.2

Two Simple Examples *Use the sign test to compare two populations for significant differences for the paired data in Exercises 4–5. State the null and alternative hypotheses to be tested. Determine an appropriate rejection region with $\alpha \le .10$. Calculate the observed value of the test statistic and present your conclusion.*

4.

	Pair						
Population	1	2	3	4	5	6	7
1	8.9	8.1	9.3	7.7	10.4	8.3	7.4
2	8.8	7.4	9.0	7.8	9.9	8.1	6.9
	+	+	+	−	+	+	+

双尾

① $H_0: p = 0.5$
$H_a: p \neq 0.5$

② $\alpha = 0.01$

③ test statistic $x = \#(+)$, $n = 7$

④ $x = 6$

$6 > 0.5 \times 7$

⑤ p-value $= 2 \cdot P(x \ge 6) = 0.125 > \alpha = 0.1$

$P(X \ge 6) = C_7^6 \cdot (0.5)^6 (0.5)^1 + C_7^7 \cdot (0.5)^7 = 0.0625$

$\boxed{\text{二項分佈 } n=7 \quad p=0.5}$

⑥ ∴ non-reject H_0 : no significant difference.

15.2

DATA SET DS1507

11. Gourmet Cooking Two chefs, A and B, rated 22 meals on a scale of 1 – 10. The data are shown in the table. Do the data provide sufficient evidence to indicate that one of the chefs tends to give higher ratings than the other? Test by using the sign test with a value of α near .05.

Meal	A	B	Meal	A	B
1	6	8	12	8	5
2	4	5	13	4	2
3	7	4	~~14~~	~~3~~	~~3~~
4	8	7	15	6	8
5	2	3	16	9	10
6	7	4	17	9	8
~~7~~	~~9~~	~~9~~	18	4	6
8	7	8	19	4	3
9	2	5	20	5	4
10	4	3	21	3	2
11	6	9	22	5	3

a. Use the binomial tables in Appendix I to find the exact rejection region for the test.

b. Use the $\boxed{\text{large-sample } z \text{ statistic.}}$ (NOTE: Although the large-sample approximation is suggested for $n \geq 25$, it works fairly well for values of n as small as 15.)

c. Compare the results of parts a and b.

(+) $x = 11$

$n = 20.$

a. $\begin{cases} H_0: & p = 0.5 \\ H_a: & p \neq 0.5 \end{cases}$ $\boxed{\text{双尾}}$ $\alpha = 0.05$

$2 \cdot P(x \leq 5) = 0.42$ 符合

$2 \cdot P(x \leq 6) = 1.16$ 不符合.

$20 - 5 = 15$

$\begin{cases} (+)\ 1, 2, 3, 4, 5 \\ (-)\ 1, 2, 3, 4, 5 \rightarrow (+)\ 15, 16, 17 \ldots 20 \text{次} \end{cases}$

$\therefore$ RR: $\{x \leq 5$ 或 $x \geq 15\}$.

$\therefore$ non-reject H_0

b. 二項分佈 ⟶ 常態分佈

$(np,\ n \times p \times (1-p))$

$\| $

$(10,\ 5)$ $\therefore$ $\mu = 10$, $\sigma = \sqrt{5} \doteq 2.236.$

$$z = \frac{x - \mu}{\sigma} = \frac{11 - 10}{2.236} = 0.447.$$

RR: $\begin{cases} \dfrac{x - 10}{2.236} < -z_{0.025} \rightarrow x = \\ x < 5.6173 \\ \dfrac{x - 10}{2.236} > z_{0.025} \\ x > 14.3827 \end{cases}$

c. ∵ 是整數 ∴ 一樣

$RR = \{x = 0, 1, 2, 3, 4, 5, 15, 16, 17, 18, 19, 20\}$

■ **Table 1 Cumulative Binomial Probabilities**
Tabulated values are $P(x \leq k) = p(0) + p(1) + \cdots + p(k)$.
(Computations are rounded to the third decimal place.)

$n = 20$

	p													
k	.01	.05	.10	.20	.30	.40	.50	.60	.70	.80	.90	.95	.99	k
0	.818	.358	.122	.012	.001	.000	.000	.000	.000	.000	.000	.000	.000	0
1	.983	.736	.392	.069	.008	.001	.000	.000	.000	.000	.000	.000	.000	1
2	.999	.925	.677	.206	.035	.004	.000	.000	.000	.000	.000	.000	.000	2
3	1.000	.984	.867	.411	.107	.016	.001	.000	.000	.000	.000	.000	.000	3
4	1.000	.997	.957	.630	.238	.051	.006	.000	.000	.000	.000	.000	.000	4
5	1.000	1.000	.989	.804	.416	.126	.021	.002	.000	.000	.000	.000	.000	5
6	1.000	1.000	.998	.913	.608	.250	.058	.006	.000	.000	.000	.000	.000	6
7	1.000	1.000	1.000	.968	.772	.416	.132	.021	.001	.000	.000	.000	.000	7
8	1.000	1.000	1.000	.990	.887	.596	.252	.057	.005	.000	.000	.000	.000	8
9	1.000	1.000	1.000	.997	.952	.755	.412	.128	.017	.001	.000	.000	.000	9
10	1.000	1.000	1.000	.999	.983	.872	.588	.245	.048	.003	.000	.000	.000	10
11	1.000	1.000	1.000	1.000	.995	.943	.748	.404	.113	.010	.000	.000	.000	11
12	1.000	1.000	1.000	1.000	.999	.979	.868	.584	.228	.032	.000	.000	.000	12
13	1.000	1.000	1.000	1.000	1.000	.994	.942	.750	.392	.087	.002	.000	.000	13
14	1.000	1.000	1.000	1.000	1.000	.998	.979	.874	.584	.196	.011	.000	.000	14
15	1.000	1.000	1.000	1.000	1.000	1.000	.994	.949	.762	.370	.043	.003	.000	15
16	1.000	1.000	1.000	1.000	1.000	1.000	.999	.984	.893	.589	.133	.016	.000	16
17	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.996	.965	.794	.323	.075	.001	17
18	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.999	.992	.931	.608	.264	.017	18
19	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.999	.988	.878	.642	.182	19
20	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	20

15.2

DATA SET DS1508

13. Recovery Rates Clinical data concerning the effectiveness of two drugs in treating a particular condition (as measured by recovery in 7 days or less) were collected from 10 hospitals. You want to know whether the data present sufficient evidence to indicate a higher recovery rate for one of the two drugs.

a. Test using the sign test. Choose your rejection region so that α is near .05.

b. Why might it be inappropriate to use the Student's t-test in analyzing the data?

	Drug A		
Hospital	Number in Group	Number Recovered (7 days or less)	Percentage Recovered
1	84	63	75.0
2	63	44	69.8
3	56	48	85.7
4	77	57	74.0
5	29	20	69.0
6	48	40	83.3
7	61	42	68.9
8	45	35	77.8
9	79	57	72.2
10	62	48	77.4

	Drug B		
Hospital	Number in Group	Number Recovered (7 days or less)	Percentage Recovered
1	96	82	85.4
2	83	69	83.1
3	91	73	80.2
4	47	35	74.5
5	60	42	70.0
6	27	22	81.5
7	69	52	75.4
8	72	57	79.2
9	89	76	85.4
10	46	37	80.4

(a)

Pair	1	2	3	4	5	6	7	8	9	10
sign	-	-	+	-	-	+	-	-	-	-

1. $H_0: p = 0.5 \quad H_a: p \neq 0.5$
2. $\alpha = 0.1$
3. test statistic: $x = \#(+), n = 10$
4. realized statistic: $x = 2$
5. $p - value = 2 * P(x \leq 2) = 2 * 0.055 = 0.11 > \alpha = 0.1$
6. Because $p - value = 0.124 > \alpha$, H_0 is not rejected, which means that there is no sufficient evidence to indicate a higher recovery rate for one of the two drugs.

(b)

If the data are not the normal distribution, the Student's t-test may give the misleading results. Therefore, it is inappropriate to use the Student's t-test in analyzing the data that are non-normal distribution.

a

n=10

1	2	3	4	5	6	7	8	9	10
-	-	+	-	-	+	-	-	-	-

① $\begin{cases} H_0 : p = 0.5 \\ H_a : p \neq 0.5. \end{cases}$

② test statistic : $\#(+), n = 10$

③ $x = 2$ $\boxed{2 < 10 \times 0.5}$

④ $p\text{-value} = 2 \times P(x \leq 2) = 2 \times (0.5^{10} + C^{10}_{1} \cdot 0.5^{10} + C^{10}_{2} \cdot 0.5^{10})$

$\fallingdotseq 2 \times 0.055 \fallingdotseq 0.11 > 0.05$ non-reject Ho

---

$p\text{-value} = 2 \cdot P(x \leq 1) = 2 \cdot 0.011 = 0.022$

$2 \cdot P(x \leq 2) = 0.11 > 0.05.$

$\therefore$ 1 & $10 - 1 = 9$.

RR: $\{x \leq 1 \text{ or } x \geq 9\}$

⑤. $x = 2$ 不在RR內

$\therefore$ non-reject Ho.

b.

t分佈：母体要常態

	n=10
	p
	.50
0	.001
1	.011
2	.055
3	.172
4	.377
5	.623
6	.828
7	.945
8	.989
9	.999
10	1.000

15.4

DATA SET DS1509

9. Refer to Exercise 4 (Section 15.2). The data in this table are from a paired-difference experiment with $n = 7$ pairs of observations.

Population	Pair 1	2	3	4	5	6	7
前 1	8.9	8.1	9.3	7.7	10.4	8.3	7.4
後. 2	8.8	7.4	9.0	7.8	9.9	8.1	6.9
d	0.1	0.7	0.3	-0.1	0.5	0.2	0.5
	(1+2)/2	7	4	(1+2)/2	(5+6)/2	3	(5+6)/2

対象

a. Use Wilcoxon's signed-rank test to determine whether there is a significant difference between the two populations.

b. Compare the results of part a with the result you got in Exercise 4 (Section 15.2). Are they the same? Explain.

An Abbreviated Version of Table 8 in Appendix I; Critical Values of T

One-Sided	Two-Sided	$n = 5$	$n = 6$	$n = 7$	$n = 8$	$n = 9$	$n = 10$	$n = 11$
$\alpha = .050$	$\alpha = .10$	1	2	4	6	8	11	14
$\alpha = .025$	$\alpha = .05$		1	2	4	6	8	11
$\alpha = .010$	$\alpha = .02$			0	2	3	5	7
$\alpha = .005$	$\alpha = .01$				0	2	3	5

One-Sided	Two-Sided	$n = 12$	$n = 13$	$n = 14$	$n = 15$	$n = 16$	$n = 17$
$\alpha = .050$	$\alpha = .10$	17	21	26	30	36	41

Two Simple Examples *Use the sign test to compare two populations for significant differences for the paired data in Exercises 4–5. State the null and alternative hypotheses to be tested. Determine an appropriate rejection region with $\alpha \le .10$. Calculate the observed value of the test statistic and present your conclusion.*

4.

Population	Pair 1	2	3	4	5	6	7
1	8.9	8.1	9.3	7.7	10.4	8.3	7.4
2	8.8	7.4	9.0	7.8	9.9	8.1	6.9

$\boxed{\text{Sign-rank}}$ $\alpha = 0.1$

① a. H_0: no significant difference
b. H_a: significant difference

②

	1	2	3	4	5	6	7
rank	$\frac{1+2}{2} = ⊕1.5$	7 ⊕	4 ⊕	$\frac{1+2}{2} = ⊖1.5$	$\frac{6+5}{2} = ⊕5.5$	3 ⊕	$\frac{6+5}{2} = ⊕5.5$

total = 28 = $\frac{(1+7)\cdot 7}{2}$ = 28 (√).

③ $n=7$, $\alpha=0.05$ $T=4$.

STAT = $\min(T^+, T^-) = \min(+26.5, \cancel{+}1.5)$
$= 1.5$

$1.5 < 4 \rightarrow$ reject H_0

$\boxed{\text{Sign}}$

① H_0: no significant difference
H_a: significant different. $\boxed{双尾}$

②

1	2	3	4	5	6	7
⊕	⊕	⊕	⊖	⊕	⊕	⊕

③ $x^* = ⊕\# = 6. \quad > 7 \cdot 0.5 = 3.5$

④ p-value $= 2 \times (P \geq x^*)$
$= 2 \times (7 \times 0.5^7 + 0.5^7)$
$= 0.125.$

$\alpha = 0.05 < 0.125$ ∴ non-reject H_0

15.4

DATA SET DS1511

11. Machine Breakdowns The number of machine breakdowns per month was recorded for 9 months on two identical machines, A and B, used to make wire rope:

pair Month	A	B	d	rank
1	3	7	− 4	6
2	14	12	+ 2	3.5
3	7	9	− 2	3.5
4	10	15	− 5	7
5	9	12	− 3	5
6	6	6	0	×
7	13	12	+ 1	1.5
8	6	5	+ 1	1.5
9	7	13	− 6	8

total = 36 = $\frac{8 \times 9}{2}$ (n)

a. Do the data provide sufficient evidence to indicate a difference in the monthly breakdown rates for the two machines? Test by using a value of α near .05.

b. Can you think of a reason the breakdown rates for the two machines might vary from month to month?

$T^- = 6+3.5+5+8+7$
$= 29.5$

1. $n = 9 - 1 = 8.$ $\quad T^+ = 3.5+1.5+1.5 = 6.5$

2. $\begin{cases} H_0: \text{difference} \\ H_a: \text{no difference} \end{cases}$ $\boxed{\text{双尾}}$

3. $\alpha = 0.05.$

4. $STAT = \min(T^+, T^-)$
$= \min(6.5, 29.5) = 6.5$

5. $T(n=8, \alpha=0.05) = 4$
$6.5 \not< 4$ ∴ non-reject H_0.

An Abbreviated Version of Table 8 in Appendix I; Critical Values of T

One-Sided	Two-Sided	$n = 5$	$n = 6$	$n = 7$	$n = 8$	$n = 9$	$n = 10$	$n = 11$
$\alpha = .050$	$\alpha = .10$	1	2	4	6	8	11	14
$\alpha = .025$	$\alpha = .05$		1	2	4	6	8	11
$\alpha = .010$	$\alpha = .02$			0	2	3	5	7
$\alpha = .005$	$\alpha = .01$				0	2	3	5

One-Sided	Two-Sided	$n = 12$	$n = 13$	$n = 14$	$n = 15$	$n = 16$	$n = 17$
$\alpha = .050$	$\alpha = .10$	17	21	26	30	36	41
$\alpha = .025$	$\alpha = .05$	14	17	21	25	30	35
$\alpha = .010$	$\alpha = .02$	10	13	16	20	24	28
$\alpha = .005$	$\alpha = .01$	7	10	13	16	19	23

15-4

15. Images and Word Recall A psychology class performed an experiment to determine whether a recall score in which instructions to form images of 25 words were given differs from an initial recall score for which no imagery instructions were given. Twenty students participated in the experiment with the results listed in the table.

DATA SET DS1514

Pair

Student	With Imagery	Without Imagery	Student	With Imagery	Without Imagery
1	20	5 +	11	17	8 +
2	24	9 +	12	20	16 +
3	20	5 +	13	20	10 +
4	18	9 +	14	16	12 +
5	22	6 +	15	24	7 +
6	19	11 +	16	22	9 +
7	20	8 +	17	25	21 +
8	19	11 +	18	21	14 +
9	17	7 +	19	19	12 +
10	21	9 +	20	23	13 +

a. What three testing procedures can be used to test for differences in the distribution of recall scores with and without imagery? What assumptions are required for the parametric procedure? Do these data satisfy these assumptions?

b. Use both the sign test and the Wilcoxon signed-rank test to test for differences in the distributions of recall scores under these two conditions.

c. Compare the results of the tests in part b. Are the conclusions the same? If not, why not?

a.

(X) pair-test

常態性 σ^2一致性. 獨立性.

(O) Rank Sum Test $\begin{cases} \text{~~常態性~~} \\ \text{~~}\sigma^2\text{一致性~~} \end{cases}$

Sign Test

只知大小無法量化 $\begin{cases} \text{~~常態性~~} \\ \text{~~}\sigma^2\text{一致性~~} \end{cases}$

Ranked - Sign Test

- pair 內差值為等距度量尺度
- pair 間可排序

→ doesn't satisfy 常態性.

b.

① $\begin{cases} H_0 : \text{no difference} \\ H_a : \text{different} \end{cases}$

② $\alpha = 0.05$

③ XSTAT = #(+).

$n = 20$

④ $x^* = 20 > 20 \times 0.5 = 10$.

⑤ p-value

$= 2 \times P(X \geq 20)$.

$= 2 \times (0.5)^{20} < 0.05$

⑥ ∴ reject H_0.

TSTAT = min.(T^+, T^-).

$T^+ = 210$

$T^- = 0$

$T = 0$.

⑤ 查表 $n = 20$ $\alpha = 0.05$ 雙尾

critical value = 52.

$0 < 52$

⑥ ∴ reject H_0.

c. yes. same result